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MIM CAPACITOR STRUCTURE WITH NON-CONFORMAL INSULATING LAYER AND FABRICATING METHOD OF THE SAME

Abstract

An MIM capacitor structure with a non-conformal insulating layer includes a substrate. A trench is disposed in the substrate. An MIM capacitor is disposed in the trench. An insulating layer non-conformally covers the trench, wherein the insulating layer contacts the MIM capacitor. The insulating layer has two overhangs formed at an opening of the trench. The overhangs seal the opening.

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Background/Summary

BACKGROUND OF THE INVENTION
1. Field of the Invention
[0001] The present invention relates to a metal-insulator-metal (MIM) capacitor, and more particularly an MIM capacitor structure with a non-conformal insulating layer.
2. Description of the Prior Art
[0002] MIM capacitors are not only used to filter noise in radio frequency circuits, or as load components in digital circuits, it is also widely used in general integrated circuit and circuit board manufacturing processes. In recent years, with the development of semiconductor integrated circuit process technology, the minimum width of devices on semiconductor substrates has gradually become smaller; therefore the density of integrated circuits per unit area is increased.
[0003] However, during the formation of MIM capacitors, the grinding powder may remain on one material layer, causing another material layer disposed on the one material layer to peel off, thereby affecting the structure of the MIM capacitor.
SUMMARY OF THE INVENTION
[0004] In view of this, the present invention provides an MIM capacitor structure with a non-conformal insulating layer to prevent grinding powder from remaining.
[0005] According to a preferred embodiment of the present invention, an MIM capacitor structure with a non-conformal insulating layer includes a substrate. A trench is disposed in the substrate. An MIM capacitor is disposed in the trench. An insulating layer non-conformally covers the trench, wherein the insulating layer contacts the MIM capacitor. The insulating layer has two overhangs formed at an opening of the trench, and the overhangs seal the opening.
[0006] According to another preferred embodiment of the present invention, a fabricating method of an MIM capacitor structure with a non-conformal insulating layer includes providing a substrate. Later, a trench is formed to be embedded in the substrate. Next, an MIM capacitor is formed to cover the trench. Finally, an insulating layer is non-conformally formed to cover the trench, and contact the MIM capacitor, wherein the insulating layer has two overhangs formed at an opening of the trench, and the overhangs seal the opening.
[0007] These and other objectives of the present invention will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiment that is illustrated in the various figures and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0008] FIG. **1** to FIG. **5** depict a fabricating method of an MIM capacitor structure with a non-conformal insulating layer according to a preferred embodiment of the present invention, wherein:
[0009] FIG. **1** depicts a substrate with a trench therein;
[0010] FIG. **2** is a fabricating stage in continuous of FIG. **1**;
[0011] FIG. **3** is a fabricating stage in continuous of FIG. **2**;

[0012] FIG. **4** is a fabricating stage in continuous of FIG. **3**; and
[0013] FIG. **5** is a fabricating stage in continuous of FIG. **4**.
[0014] FIG. **6** to FIG. **8** depict a fabricating method of an MIM capacitor structure according to an examplary embodiemnt of the present invention, wherein:
[0015] FIG. **6** depicts a recess on a trench;
[0016] FIG. **7** is a fabricating stage in continuous of FIG. **6**; and
[0017] FIG. **8** is a fabricating stage in continuous of FIG. **7**.

DETAILED DESCRIPTION

[0018] FIG. **1** to FIG. **5** depict a fabricating method of an MIM capacitor structure with a non-conformal insulating layer according to a preferred embodiment of the present invention.
[0019] As shown in FIG. **1**, a substrate **10** is provided. Then, at least one trench **12** is formed to embed in the substrate **10**. In FIG. **1**, two trenches **12** are formed for an example, but the number of the trench **12** is not limited to two. Each of the trenches **12** includes a bottom **12***a* and a sidewall **12***b*. The bottom **12***a* and the sidewall **12***b* form an inner corner A. It is noteworthy that the angle of the inner corner A is between 85 and 90 degrees (excluding 90 degrees). In this way, the width of the trench **12** is reduced from the bottom **12***a* to the opening of the trench **12**, which helps a non-conformal insulating layer to seal the opening of the trench **12** later.
[0020] As shown in FIG. **2**, a silicon oxide liner **14** is formed to conformally cover and contact the trench **12** and cover the top surface of the substrate **10**. Later, a first electrode E**1**, a capacitor dielectric layer I and a second electrode E**2** are formed in sequence to cover and contact the silicon oxide liner **14**. The first electrode E**1**, the capacitor dielectric layer I and the second electrode E**2** together form an MIM capacitor C**1**. In this embodiment, the first electrode E**1**, the capacitor dielectric layer I and the second electrode E**2** continuously fill the two trenches **12**. As shown in FIG. **3**, a non-conformal insulating layer **16** is formed to cover the MIM capacitor C**1**. “Non-conformal” is because the thickness of the insulating layer **16** is inconsistent, the profile of the insulating layer **16** can't follow the profile of the trench **12**. Specifically speaking, the thickness of the insulating layer **16** disposed at the inner corner A of the trench **12** (please refer to FIG. **1**) is greater than the thickness of the insulating layer **16** disposed on the sidewall **12***b* of the trench **12**. The thickness of the insulating layer **16** disposed at the opening of the trench **12** is greater than the thickness of the insulating layer **16** disposed on the sidewall **12***b* of the trench **12**. In this way, two overhangs **16***a*/**16***b* of the insulating layer **16** are formed at the opening of the trench **12**, and the overhangs **16***a*/**16***b* together seal the opening. An air gap AG is formed in the insulating layer **16**. The insulating layer **16** may be formed by a chemical vapor deposition or a physical vapor deposition. In addition, a recess **18** is formed at the interface of the two overhangs **16***a*/**16***b*. The insulating layer **16** includes silicon nitride, silicon oxynitride, silicon nitride carbide or silicon oxide.
[0021] As shown in FIG. **4**, a silicon oxide layer **20** is formed to cover the insulating layer **16**, and the silicon oxide layer **20** fills the recess **18**. As shown in FIG. **5**, after planarizing the silicon oxide layer **20**, part of the silicon oxide layer **20** and part of the insulating layer **16** on the top surface of the substrate **10** are removed. Later, part of the first electrode E**1**, part of the capacitor dielectric layer I and part of the second electrode E**2** are removed. Next, a silicon nitride layer **22** is formed to cover the silicon oxide layer **20**, the second electrode E**2** and the first electrode E**1**. Subseqenctly, a dielectric layer **24** is formed to cover the silicon nitride layer **22** and the MIM capacitor C**1**. Then, a first contact plug **26***a* and a second contact plug **26***b* are formed to penetrate the dielectric layer **24** and the silicon nitride layer **22**. The first contact plug **26***a* contacts the first electrode E**1** and the second contact plug **26***b* contacts the second electrode E**2**. Now, an MIM capacitor structure **100** with a non-conformal insulating layer of the present invention is completed.
[0022] As shown in FIG. **5**, an MIM capacitor structure **100** with a non-conformal insulating layer includes a substrate **10**. The substrate **10** includes a silicon substrate, a germanium substrate, a gallium arsenide substrate, a silicon germanium substrate, an indium phosphide substrate, a gallium

nitride substrate, a silicon carbide substrate or a silicon on insulator substrate. A trench **12** is disposed in the substrate **10**. A silicon oxide liner **14** covers and contacts the trench **12**. An MIM capacitor C**1** is disposed in the trench **12** and contacts the silicon oxide liner **14**. An insulating layer **16** non-conformally covers the trench **12**, and the insulating layer **16** contacts the MIM capacitor C**1**. The insulating layer **16** forms two overhangs **16***a*/**16***b* at an opening of the trench **12**, and two overhangs **16***a*/**16***b* seal the opening. The insulating layer **16** includes silicon nitride, silicon oxynitride, silicon nitride carbide or silicon oxide.

[0023] The MIM capacitor C**1** includes a first electrode E**1**, a capacitor dielectric layer I and a second electrode E**2**. An air gap AG is disposed in the insulating layer **16** and located within the trench **12**. The first electrode E**1** and the second electrode E**2** respectively include tantalum nitride, titanium nitride, tantalum, titanium, aluminum or poly-crystalline silicon. The capacitor dielectric layer I includes aluminum oxide, zirconium oxide, barium strontium titanate (BST), lead zirconate titanate (PZT), zirconium silicate (ZrSiO.sub.4), hafnium silicon oxide (HfSiO.sub.2), hafnium silicon oxynitride (HfSiON), tantalum oxide or a combination of the above materials.

[0024] Moreover, the trench **12** includes a bottom **12***a* and a sidewall **12***b*. The bottom **12***a* and the sidewall **12***b* form an inner angle A (please refer to FIG. **1**). The inner angle A faces the MIM capacitor C**1**, and the angle of the inner angle A is between 85 and 90 degrees. The thickness of the insulating layer **16** disposed at the inner corner A is greater than the thickness of the insulating layer **16** disposed on the sidewall **12***b* of the trench **12**, and the thickness of the insulating layer **16** disposed at the opening of the trench **12** is greater than the thickness of the insulating layer **16** disposed on the sidewall **12***b* of the trench **12**. The insulating layer **16** located at the opening of the trench **12** is defined as overhangs **16***a*/**16***b*. In addition, the silicon oxide layer **20** covers and contacts the insulating layer **16**. A silicon nitride layer **22** covers the silicon oxide layer **20**, the second electrode E**2** and the first electrode E**1**.

[0025] FIG. **6** to FIG. **8** depict a fabricating method of an MIM capacitor structure according to an examplary embodiemnt of the present invention, wherein like reference numerals are used to refer to like elements throughout.

[0026] As shown in FIG. **6**, a substrate **10** is provided. Then, at least one trench **112** is formed in the substrate **10**. The trench **112** includes a bottom **112***a* and a sidewall **112***b*. The bottom **112***a* and the sidewall **112***b* form an inner angle B. The inner angle is 90 degrees or greater than 90 degrees. In this way, the width of the bottom **112***a* of the trench **112** may be equal to or smaller than the width of the opening of the trench **112**. Therefore, air gaps are less likely to form in trench **112**. Thereafter, a silicon oxide liner **14**, a first electrode E**1**, a capacitor dielectric layer I and a second electrode E**2** are sequentially formed to cover the trench **112** and the top surface of the substrate **10**. The first electrode E**1**, the capacitor dielectric layer I and the second electrode E**2** together form an MIM capacitor C**2**. Later, a silicon oxide layer **120** is formed to cover the second electrode E**2** and fill in the trench **112**. The silicon oxide layer **120**, the first electrode E**1**, the capacitor dielectric layer I and the second electrode E**2** together fill up the trench **112**. The silicon oxide layer **120** needs to fill the space in the trench **112** where there is no the first electrode E**1**, the capacitor dielectric layer I or the second electrode E**2** filling in, therefore the silicon oxide layer **120** will form a recess **30** on the opening of the trench **112**. The lowest point of the recess **30** is sandwiched between the second electrode E**2** disposed on the two sidewalls **112***b* of the trench **112**. The silicon oxide layer **120** may be formed by a chemical vapor deposition, a physical vapor deposition, or an atomic layer deposition. In this embodiment, the silicon oxide layer **120** is preferably formed by an atomic layer deposition.

[0027] As shown in FIG. **7**, the silicon oxide layer **120** is planarized. The method of planarizing the silicon oxide layer **120** is preferably performed by a chemical mechanical polishing process. Since slurry is used during the chemical mechanical polishing process, the grinding powder **32** in the slurry may stuck in the recess **30**. Because the lowest point of the recess **30** is sandwiched between the second electrode E**2** located on the two sidewalls **112***b* of the trench **112**, the recess **30** of the

silicon oxide layer **120** cannot be completely removed, otherwise the second electrode E**2** will be ground. In order to protect the second electrode E**2** during the chemical mechanical polishing process, a certain thickness of the silicon oxide layer **120** needs to be maitained on the second electrode E**2**. As a result, after the chemical mechanical polishing process is completed, part of the recess **30** still remains, and the grinding powder **32** remains in the recess **30**.
[0028] As shown in FIG. **8**, part of the silicon oxide layer **120** on the top surface of the substrate **10** is removed. The subsequent processes are the same as those in FIG. **5**. For example, part of the silicon oxide layer **120**, part of the first electrode E**1**, part of the capacitor dielectric layer I and part of the second electrode E**2** are removed. Then, a silicon nitride layer **22** is formed to cover the silicon oxide layer **120**, the recess **30**, the second electrode E**2** and the first electrode E**1**. Later, the dielectric layer **24**, the first contact plug **26***a* and the second contact plug **26***b* are formed. Now, an MIM capacitor structure **200** of the present invention is completed.
[0029] In the MIM capacitor structure **200**, there is grinding powder **32** remaining between the silicon nitride layer **22** and the silicon oxide layer **120**, therefore the silicon nitride layer **22** can not attach to the silicon oxide layer **120** completely. In this way, the silicon nitride layer **22** is easily delaminated, damaging the MIM capacitor structure **200**. Regarding the MIM capacitor structure **100** with a non-conformal insulating layer, the opening of the trench **12** is seal up by the non-conformal insulating layer **16**, and the thickness of the insulating layer **16** is smaller than that of the silicon oxide layer **120**. Therefore, the depth of the recess **18** of the insulating layer **16** is smaller than the depth of the recess **30** of the silicon oxide layer **120**. That is, the recess **18** is flatter than the recess **30**. By this process, when the silicon oxide layer **20** is formed on the insulating layer **18** thereafter, the silicon oxide layer **20** can fill up the recess **18**. Later, the silicon oxide layer **20** is planarized. Comparing to the examplary embodiment, the process in the preferred embodiment does not have the recess on the silicon oxide layer **20**, and without grinding powder **32** remained. Therefore, in the MIM capacitor structure **100** with a non-conformal insulating layer, the silicon nitride layer **22** can completely attach to the silicon oxide layer **20**.
[0030] Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

Claims

1. A metal-insulator-metal (MIM) capacitor structure with a non-conformal insulating layer, comprising: a substrate; a trench disposed in the substrate; an MIM capacitor disposed in the trench; and an insulating layer non-conformally covering the trench, wherein the insulating layer contacts the MIM capacitor, the insulating layer has two overhangs formed at an opening of the trench, and the overhangs seal the opening.
2. The MIM capacitor structure with a non-conformal insulating layer of claim 1, further comprising a silicon oxide layer covering and contacting the insulating layer.
3. The MIM capacitor structure with a non-conformal insulating layer of claim 1, wherein the trench comprises a bottom and a sidewall, the bottom and the sidewall form an inner corner, the inner corner faces the MIM capacitor, and an angle of the inner corner is between 85 and 90 degrees.
4. The MIM capacitor structure with a non-conformal insulating layer of claim 3, wherein a thickness of the insulating layer at the inner corner is greater than a thickness of the insulating layer on the sidewall of the trench.
5. The MIM capacitor structure with a non-conformal insulating layer of claim 1, wherein a thickness of the insulating layer disposed at the opening is greater than a thickness of the insulating layer disposed at a sidewall of the trench.

6. The MIM capacitor structure with a non-conformal insulating layer of claim 1, further comprising a silicon oxide liner covering and contacting the trench, and the MIM capacitor contacting the silicon oxide liner.
7. The MIM capacitor structure with a non-conformal insulating layer of claim 1, wherein an air gap is disposed in the insulating layer.
8. The MIM capacitor structure with a non-conformal insulating layer of claim 1, wherein the insulating layer comprises silicon nitride, silicon oxynitride, silicon carbon nitride or silicon oxide.
9. A fabricating method of a metal-insulator-metal (MIM) capacitor structure with a non-conformal insulating layer, comprising: providing a substrate; forming a trench embedded in the substrate; forming an MIM capacitor covering the trench; and non-conformally forming an insulating layer covering the trench, and contacting the MIM capacitor, wherein the insulating layer has two overhangs formed at an opening of the trench, and the overhangs seal the opening.
10. The fabricating method of an MIM capacitor structure with a non-conformal insulating layer of claim 9, further comprising: forming a silicon oxide layer covering and contacting the insulating layer; and planarizing the silicon oxide layer.
11. The fabricating method of an MIM capacitor structure with a non-conformal insulating layer of claim 9, wherein the trench comprises a bottom and a sidewall, the bottom and the sidewall form an inner corner, the inner corner faces the MIM capacitor, and an angle of the inner corner is between 85 and 90 degrees.
12. The fabricating method of an MIM capacitor structure with a non-conformal insulating layer of claim 11, wherein a thickness of the insulating layer at the inner corner is greater than a thickness of the insulating layer on the sidewall of the trench.
13. The fabricating method of an MIM capacitor structure with a non-conformal insulating layer of claim 9, wherein a thickness of the insulating layer disposed at the opening is greater than a thickness of the insulating layer disposed at a sidewall of the trench.
14. The fabricating method of an MIM capacitor structure with a non-conformal insulating layer of claim 9, wherein an air gap is disposed in the insulating layer.
15. The fabricating method of an MIM capacitor structure with a non-conformal insulating layer of claim 9, wherein the insulating layer comprises silicon nitride, silicon oxynitride, silicon carbon nitride or silicon oxide.